

DATA 308: Lab 5b

Hypothesis Testing with SLR

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Our research question that we will investigate in this lab is:

Among first-year Elmhurst College students represented by these data, is family income linearly associated with mean gift aid received?

```
library(openintro)
library(broom)
library(tidyverse)
data("elmhurst")
```

Question 1

Take a look at the `elmhurst` dataset and familiarize yourself with it. You can run the code chunk below to view help documentation for this dataset and the structure. I have the `eval` argument chunk set to false, meaning when you render this document the code chunk will not be run.

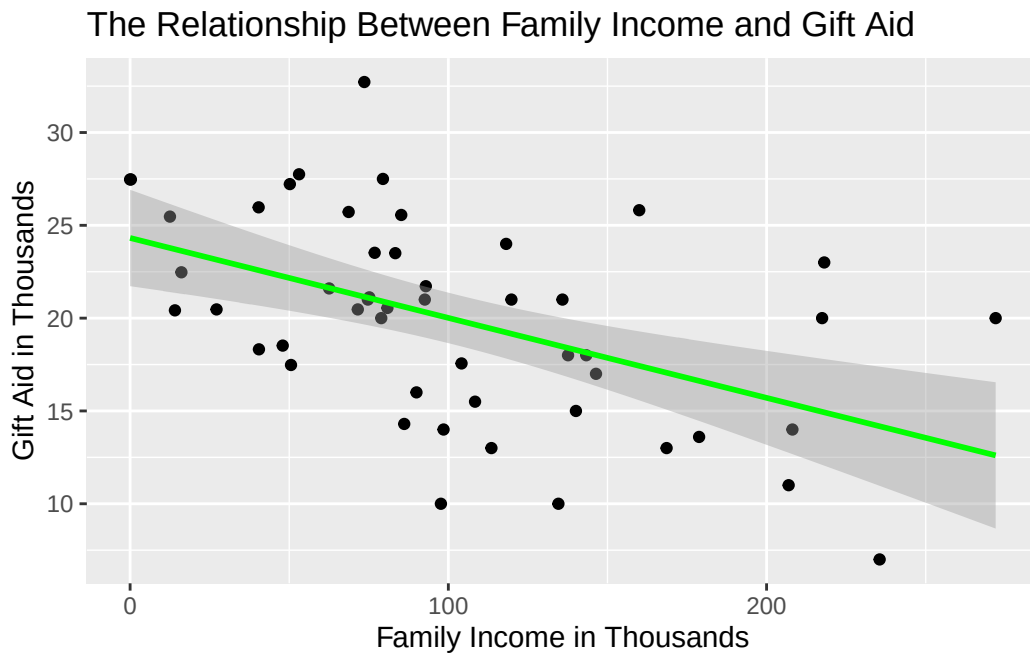
```
?elmhurst
glimpse(elmhurst)
```

Question 2

Produce a plot to visualize the relationship between gift aid and family income. Make sure to plot the response variable on the y-axis and include a regression line.

Response

```
ggplot(elmhurst, aes(x = family_income, y = gift_aid)) +
  geom_point() +
  geom_smooth(method = "lm", color = "green") +
  labs(
    x = "Family Income in Thousands",
    y = "Gift Aid in Thousands",
    title = "The Relationship Between Family Income and Gift Aid"
  )
)
```



Question 3

We will learn about the conditions for linear regression next week, but for now just visually assess, does a line appear to be a good model?

Response

It doesn't appear to be a particularly good model, but it does seem to inform us that there's a weak and negative relationship between `family_income` and `gift_aid`.

Question 4

Below is the population simple linear regression model, in a general form written with LaTeX. Replace the Y and X in the equation with the response and explanatory variables for this context.

Response

$$\text{Gift Aid Received} = \beta_0 + \beta_1(\text{Family Income}) + \epsilon$$

Question 5

State the null and alternative hypotheses, in symbols and in words, to test our research question stated at the top of this document.

Response

There is no linear association between family income and the amount of gift aid a student receives at Elmhurst college in 2011.

$$H_0 = \beta_1 = 0$$

There is a linear association between family income and the amount of gift aid a student receives at Elmhurst college in 2011.

$$H_a = \beta_1 \neq 0$$

Question 6

Fit a simple linear regression model.

Response

```
aid_model <- lm(gift_aid ~ family_income, data = elmhurst)
tidy_model <- tidy(aid_model)

summary(aid_model)
```

```

Call:
lm(formula = gift_aid ~ family_income, data = elmhurst)

Residuals:
    Min       1Q   Median       3Q      Max
-10.1128  -3.6234  -0.2161   3.1587  11.5707

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  24.31933     1.29145   18.831 < 2e-16 ***
family_income -0.04307     0.01081   -3.985 0.000229 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.783 on 48 degrees of freedom
Multiple R-squared:  0.2486,    Adjusted R-squared:  0.2329
F-statistic: 15.88 on 1 and 48 DF,  p-value: 0.0002289

```

Question 7

What is the p-value for this hypothesis test? With a significance level of 0.05, what is your statistical decision?

Response

```

p_value <- tidy_model |>
  filter(term == "family_income") |>
  select(p.value)

p_value

```

```

# A tibble: 1 x 1
  p.value
  <dbl>
1 0.000229

```

$$p = 0.0002289$$

$$p < 0.05$$

Our p-value ($p = 0.00023$) is less than our significance level, so we can reject the null hypothesis ($H_0 = 0$).

Question 8

We should not just report a p-value on its own, we should also provide context. Compute a confidence interval for β_1 .

Response

```
aid_ci <- confint(aid_model)

aid_ci
```

	2.5 %	97.5 %
(Intercept)	21.72269421	26.91596380
family_income	-0.06480555	-0.02133775

Question 9

Provide your conclusion in context for this hypothesis test. One sentence should explain the results of your hypothesis test, with p-value reported in parentheses. The next sentence should provide your estimate and confidence interval interpreted in context (you can choose to report it as an interval or as an margin of error).

Response

We found strong evidence of a linear relationship ($p = 0.00023$) between the amount of gift aid received and family income at Elmhurst college in 2011. We estimate an increase in family income of \$1,000 to be associated with a decrease in mean gift aid received of \$43, between \$21 and \$64 with 95% confidence.

Final note

A condition on our conclusion:

Our hypothesis test is valid only if certain conditions about the data and regression model are satisfied. Next week, we will learn how to identify these conditions and use diagnostic plots to assess them. For now, our conclusion assumes that the required conditions hold.